# Randomized Median Selection

## A Sampling-Based Approach

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# Outline


1. Problem and Motivation
2. Sampling Strategy and the Algorithm
3. Correctness and Linear-Time Termination
4. Failure Events and Their Probabilities
5. Takeaways

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# Learning goals

- Understand the sampling-based median algorithm.
- See why one run always takes linear time.
- Prove that one run fails with probability at most $n^{-1/4}$.
- Interpret the algorithm as a Monte Carlo algorithm, and see how repetition gives a Las Vegas algorithm.

# Median selection problem

Given a set $S$ of $n$ distinct elements from a totally ordered universe, a median is an element $m \in S$ such that

$$\#\{x \in S : x \leq m\} \geq \lceil n/2 \rceil \quad \text{and} \quad \#\{x \in S : x \geq m\} \geq \lceil n/2 \rceil.$$

Consider the simplifying assumptions:

- $n$ is odd;
- all elements of $S$ are distinct.

Hence $m$ is exactly the $\left(\frac{n+1}{2}\right)$th element in sorted order.

# Deterministic baselines

## Two classical approaches

- Sort all of $S$: $O(n \log n)$ time.
- A deterministic $O(n)$ time selection algorithm, but more complicated. [Blum, Floyd, Pratt, Rivest, Tarjan; JCSS 1973]

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## Our Objective

Design a *simpler* randomized algorithm that still runs in linear time per attempt and succeeds with high probability.

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2. the interval
$$C := \{x \in S : d \leq x \leq u\}$$
is small enough to sort quickly.

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2. the interval
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is small enough to sort quickly.

If we can find such a narrow interval around the true median, then the remaining work is easy.

# Another intuitive idea

Suppose we randomly select two numbers $a$ and $b$, $a \leq b$, from a list $N$ of $n$ numbers (with replacement), what is the probability that for we have the median of $N$ is not between $a$ and $b$?

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Suppose we randomly select three numbers $a$, $b$, and $c$, $a \leq b \leq c$, from a list $N$ of $n$ numbers (with replacement), what is the probability that for we have the median of $N$ is not between $a$ and $c$?

# Desired sandwich interval

We want

$$d \leq m \leq u \quad \text{and} \quad |C| = O(n^{3/4}).$$

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**Key idea**

Shrink the search space from all of $S$ down to a small candidate interval $C$.

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**Main challenge**

How to find $d$ and $u$ quickly, without knowing the median?

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**Why with replacement?**

- easier to implement;
- easier to analyze because the samples are independent.

# How $d$ and $u$ are chosen?

After sorting $R$, define:

$$d = \text{the } \left(\frac{1}{2}n^{3/4} - \sqrt{n}\right)\text{th smallest element of } R,$$

$$u = \text{the } \left(\frac{1}{2}n^{3/4} + \sqrt{n}\right)\text{th smallest element of } R.$$

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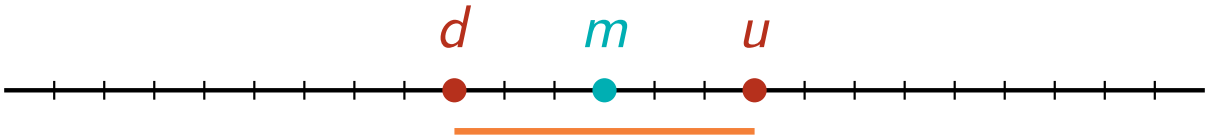
$d$ and $u$ are symmetrically around the sample median, with a buffer of $\sqrt{n}$ sample positions on each side.

- This buffer is what makes the true median fall inside $[d, u]$ *with high probability*.




$$C = \{x \in S : d \leq x \leq u\}$$

Goal: $m$ lies in a "short" interval that can be sorted cheaply.

- $d$ and $u$ are selected from the sample $R$.
- $C$ is defined in the original set $S$.

# Algorithm: Randomized Median Algorithm (RMA)

**Input / output**

Input: a set $S$ of $n$ elements over a totally ordered universe.
Output: the median element of $S$, denoted by $m$.

1. Pick a multiset $R$ of $n^{3/4}$ elements from $S$, independently and uniformly at random with replacement.
2. Sort $R$.
3. Let $d$ be the $\left(\frac{1}{2}n^{3/4} - \sqrt{n}\right)$th smallest element of $R$.
4. Let $u$ be the $\left(\frac{1}{2}n^{3/4} + \sqrt{n}\right)$th smallest element of $R$.
5. Compare every element of $S$ with $d$ and $u$ to compute
   $$C = \{x \in S : d \le x \le u\}, \quad \ell_d = \#\{x \in S : x < d\}, \quad \ell_u = \#\{x \in S : x > u\}.$$
6. If $\ell_d > n/2$ or $\ell_u > n/2$, output $\textsc{Fail}$.
7. If $|C| \le 4n^{3/4}$, sort $C$; otherwise output $\textsc{Fail}$.
8. Output the $\left(\lceil n/2 \rceil - \ell_d + 1\right)$th smallest element of $C$.

# Purposes of each step

| Step | Purpose |
|---|---|
| 1–4 | Use a random sample to guess a narrow interval around the true median. |
| 5 | Translate the sample interval $[d, u]$ back to the full set $S$. |
| 6 | Check whether the true median could have escaped outside $[d, u]$. |
| 7 | Guarantee the sorting work remains sublinear. |
| 8 | Recover the exact rank of the median inside $C$. |

# Numerical scale of the parameters

Take $n = 10^4$ as a sanity check.

$$n^{3/4} = 1{,}000, \qquad \sqrt{n} = 100.$$

So the algorithm:

- samples only 1,000 items from 10,000;
- chooses $d$ and $u$ at sample ranks $500 - 100 = 400$ and $500 + 100 = 600$;
- accepts only if $|C| \leq 4n^{3/4} = 4{,}000$.

We pay linear cost to isolate a candidate interval that is much smaller than the whole input and likely to contain the true median.

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# The Main Theorem

**Theorem**

*Algorithm RMA terminates in linear time, and if it does not output* FAIL, *then it outputs the correct median element of the input set $S$.*

This theorem separates the analysis into two parts:

1. **deterministic correctness conditioned on no failure**;
2. **probability of failure**.

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Therefore, if $m \notin C$, step 6 forces the algorithm to output FAIL.

So if the algorithm reaches step 8, the true median must already lie in $C$.

# Why step 8 returns the median once $m \in C$

Assume $m \in C$.

- There are exactly $\ell_d$ elements of $S$ smaller than $d$, and all remaining elements smaller than $m$ that are not among those $\ell_d$ lie inside $C$.
- Hence inside the sorted order of $C$, the median sits at position

$$\lceil n/2 \rceil - \ell_d + 1.$$

Under the event $m \in C$, step 8 returns the exact median of $S$.

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If $|C|$ is too large, step 7 stops immediately with FAIL instead of spending superlinear time.

# Deterministic summary so far

**What is already proved without probability?**

1. Every execution halts in $O(n)$ time.
2. Any non-FAIL output is correct.

**What remains?**

Bound the probability that the algorithm outputs FAIL.

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# Three bad events

Define

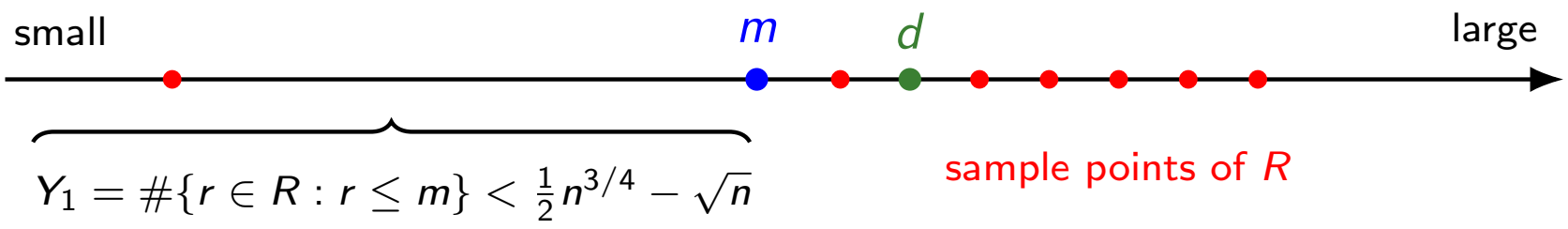
$$E_1 : \; Y_1 := \#\{r \in R : r \leq m\} < \frac{1}{2} n^{3/4} - \sqrt{n},$$
$$E_2 : \; Y_2 := \#\{r \in R : r \geq m\} < \frac{1}{2} n^{3/4} - \sqrt{n},$$
$$E_3 : \; |C| > 4n^{3/4}.$$

Intuitive meanings:

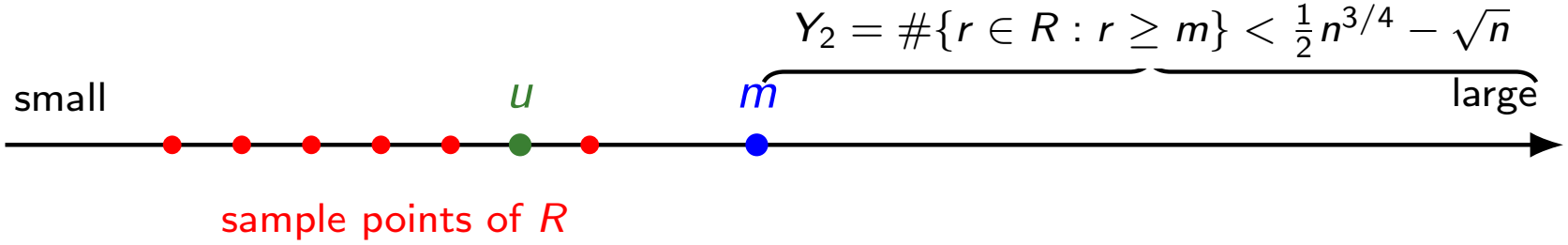
- $E_1$: too few sampled points land at or below the true median;
- $E_2$: too few sampled points land at or above the true median;
- $E_3$: the candidate interval in the full set is still too large.

# Bad events $E_1$ and $E_2$: picture intuition

$E_1$: too few sampled points land at or below $m$




$E_2$: too few sampled points land at or above $m$



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Bound $\Pr[E_1]$, $\Pr[E_2]$, and $\Pr[E_3]$ separately, then apply the union bound.

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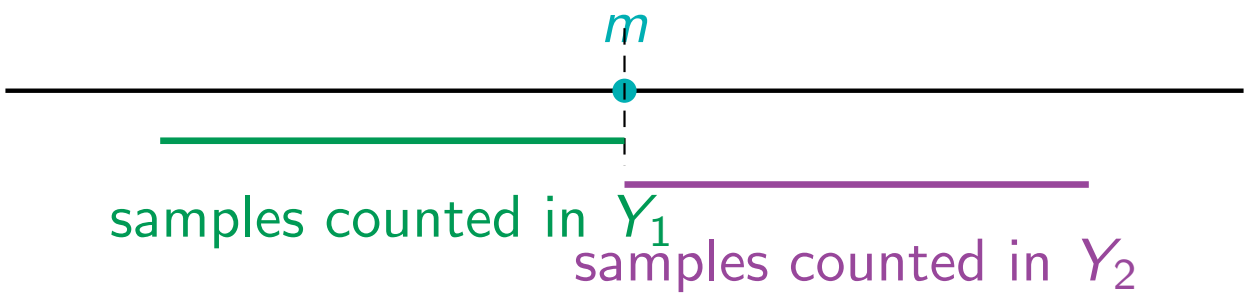
**Proof Idea**

Bound $\Pr[E_1]$, $\Pr[E_2]$, and $\Pr[E_3]$ separately, then apply the union bound.

$$\Pr[E_1 \cup E_2 \cup E_3] \leq \Pr[E_1] + \Pr[E_2] + \Pr[E_3].$$

# Visual intuition for $E_1$ and $E_2$




If $Y_1$ is too small, then $d$ ends up to the *right* of $m$.
If $Y_2$ is too small, then $u$ ends up to the *left* of $m$.

# Bounding $\Pr[E_1]$: define indicators

For each sampled point $r_i \in R$, define

$$X_i = \begin{cases} 1, & r_i \leq m, \\ 0, & r_i > m. \end{cases}$$

Then

$$Y_1 = \sum_{i=1}^{n^{3/4}} X_i.$$

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Because the sample is taken with replacement, $X_1, X_2, \ldots, X_{n^{3/4}}$ are independent Bernoulli trials.

# Expectation of $Y_1$

Since $m$ is the median and all elements are distinct,

$$\Pr[X_i = 1] = \frac{(n-1)/2 + 1}{n} = \frac{1}{2} + \frac{1}{2n}.$$

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Therefore

$$\mathbb{E}[Y_1] = n^{3/4}\left(\frac{1}{2} + \frac{1}{2n}\right) = \frac{1}{2}n^{3/4} + \frac{1}{2n^{1/4}}.$$

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## Comparison with the bad threshold

Event $E_1$ asks whether $Y_1$ drops below

$$\frac{1}{2}n^{3/4} - \sqrt{n},$$

which is roughly $\sqrt{n}$ below its mean.

# Variance of $Y_1$

- Because $Y_1$ is binomial with parameters $n^{3/4}$ and $\frac{1}{2} + \frac{1}{2n}$,

$$\mathrm{Var}[Y_1] = n^{3/4}\left(\frac{1}{2} + \frac{1}{2n}\right)\left(\frac{1}{2} - \frac{1}{2n}\right) = \frac{1}{4}n^{3/4} - \frac{1}{4n^{5/4}} < \frac{1}{4}n^{3/4}.$$

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- So the standard deviation scale is about $n^{3/8}$, while the buffer in the algorithm is $\sqrt{n} = n^{1/2}$.

# Chebyshev bound for $E_1$

Using Chebyshev's inequality,

$$\begin{aligned}
\Pr[E_1] &= \Pr\left[Y_1 < \frac{1}{2}n^{3/4} - \sqrt{n}\right] \\
&\leq \Pr\left[|Y_1 - \mathbb{E}[Y_1]| > \sqrt{n}\right] \\
&\leq \frac{\mathrm{Var}[Y_1]}{(\sqrt{n})^2} \\
&< \frac{\frac{1}{4}n^{3/4}}{n} = \frac{1}{4}n^{-1/4}.
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**Lemma**

$$\Pr[E_1] \leq \frac{1}{4}n^{-1/4}.$$

# The bound for $E_2$

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## Cumulative progress

So far,

$$\Pr[E_1] + \Pr[E_2] \leq \frac{1}{2} n^{-1/4}.$$

# Why $E_3$ is harder

Event $E_3$ is different:

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Event $E_3$ is different:

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- Even if the true median lies inside $[d, u]$, the interval could still be too wide.
- We must show that the chosen sample ranks make the resulting interval in $S$ *unlikely to contain too many elements*.

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- If $|C| > 4n^{3/4}$, then at least one of the following must happen:

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- If $|C| > 4n^{3/4}$, then at least one of the following must happen:

  $E_{3,1}$ : at least $2n^{3/4}$ elements of $C$ are **greater** than $m$,

  $E_{3,2}$ : at least $2n^{3/4}$ elements of $C$ are **smaller** than $m$.

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  $E_{3,1}$ : at least $2n^{3/4}$ elements of $C$ are **greater** than $m$,

  $E_{3,2}$ : at least $2n^{3/4}$ elements of $C$ are **smaller** than $m$.

- Thus,

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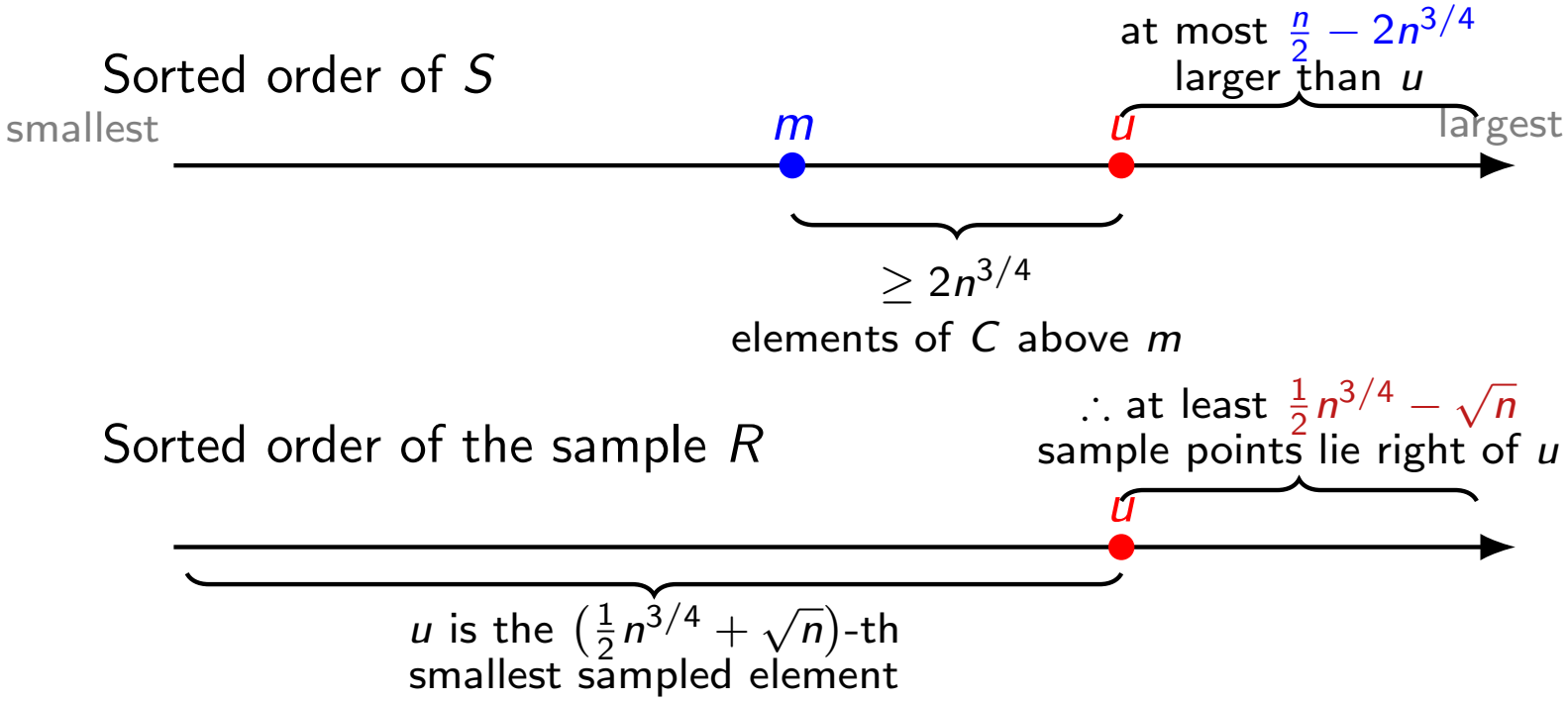
- By symmetry, it is enough to bound $\Pr[E_{3,1}]$.

# What does $E_{3,1}$ imply? (1/3)

$E_{3,1}$: at least $2n^{3/4}$ elements of $C$ are **greater** than $m$.

- If $E_{3,1}$ occurs, then $u$ is too far above the median in the sorted order of $S$.

# What does $E_{3,1}$ imply? (2/3)




$$E_{3,1} \implies \text{rank}_S(u) \geq \frac{n}{2} + 2n^{3/4}.$$

# What does $E_{3,1}$ imply? (3/3)

- The rank of $u$ in $S$ is at least

$$\frac{1}{2}n + 2n^{3/4}.$$

- Among the sample points in $R$, at least $\frac{1}{2}n^{3/4} - \sqrt{n}$ samples must come from the $\frac{1}{2}n - 2n^{3/4}$ largest elements of $S$.

# Define the variable for $E_{3,1}$

- Let $X$ be the number of sampled points falling among the largest

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- Then $E_{3,1}$ implies $X \geq \frac{1}{2}n^{3/4} - \sqrt{n}$.

# Expectation and variance of $X$

The success probability for each $X_i$ is

$$\frac{\frac{1}{2}n - 2n^{3/4}}{n} = \frac{1}{2} - 2n^{-1/4}.$$

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Also,

$$\begin{aligned}\mathrm{Var}[X] &= n^{3/4}\left(\frac{1}{2} - 2n^{-1/4}\right)\left(\frac{1}{2} + 2n^{-1/4}\right) \\ &= \frac{1}{4}n^{3/4} - 4n^{1/4} < \frac{1}{4}n^{3/4}.\end{aligned}$$

# Chebyshev bound for $E_{3,1}$

Now apply Chebyshev again:

$$\begin{aligned}
\Pr[E_{3,1}] &\le \Pr\left[X \ge \frac{1}{2}n^{3/4} - \sqrt{n}\right] \\
&\le \Pr\left[|X - \mathbb{E}[X]| \ge \sqrt{n}\right] \\
&\le \frac{\mathrm{Var}[X]}{(\sqrt{n})^2} \\
&< \frac{\frac{1}{4}n^{3/4}}{n} = \frac{1}{4}n^{-1/4}.
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By symmetry, $\Pr[E_{3,2}] \le \frac{1}{4}n^{-1/4}$. Therefore, $\Pr[E_3] \le \frac{1}{2}n^{-1/4}$.

# Finally, using the union bound

Combining the three bounds,

$$\Pr[E_1] + \Pr[E_2] + \Pr[E_3] \leq \frac{1}{4}n^{-1/4} + \frac{1}{4}n^{-1/4} + \frac{1}{2}n^{-1/4} = n^{-1/4}.$$

**Theorem**

*The probability that Algorithm RMA fails is at most $n^{-1/4}$.*

**Equivalent success guarantee**

A single run succeeds with probability at least

$$1 - n^{-1/4}.$$

# What was the role of Chebyshev's inequality?

For each bad event, we:

1. express a sample count as a sum of independent Bernoulli indicators;
2. compute its mean and variance;
3. use Chebyshev to control deviations of size $\sqrt{n}$.

The algorithm is a concrete application of variance calculations and Chebyshev bounds, not yet of Chernoff bounds.

# Outline

1 Problem and Motivation

2 Sampling Strategy and the Algorithm

3 Correctness and Linear-Time Termination

4 Failure Events and Their Probabilities

5 Takeaways

# One-run guarantee

For one execution:

- running time is always $O(n)$;
- output is either the correct median or FAIL;
- failure probability is at most $n^{-1/4}$.

This is a *Monte Carlo* algorithm: bounded running time, but a small probability of failure.

# From Monte Carlo to Las Vegas

Repeat Algorithm RMA until it succeeds.

Because the samples used in different runs are independent,

- each run succeeds with probability at least $1 - n^{-1/4}$;
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The repeated version always returns the correct median.

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**Result**

The repeated version always returns the correct median.

So repetition converts the Monte Carlo algorithm into a *Las Vegas* algorithm.

# Expected running time after repetition

Let $p \geq 1 - n^{-1/4}$ be the success probability of one run. For a geometric random variable with success probability $p$,

$$\mathbb{E}[\text{number of runs}] = \frac{1}{p}.$$

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Hence for sufficiently large $n$, this is still $O(n)$.

**Conclusion**

The repeated algorithm is Las Vegas with linear expected running time.

# Why these parameters are balanced

There are three competing design goals:

1. sample small enough so sorting $R$ is cheap;
2. buffer large enough so $m$ lies between $d$ and $u$ with high probability;
3. candidate set $C$ small enough so sorting $C$ is cheap.

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There are three competing design goals:

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The algorithm chooses

$$|R| = n^{3/4}, \quad \text{buffer} = \sqrt{n}, \quad \text{accept if } |C| \leq 4n^{3/4},$$

which makes all three constraints fit together cleanly with Chebyshev bounds.

# Big-picture summary

1. Sample $n^{3/4}$ elements.
2. Use the sample to define a narrow interval $[d, u]$ around the sample median.
3. Scan the original input to form the candidate set $C$.
4. Abort if $C$ is too large or if the median appears to fall outside the interval.
5. Otherwise sort only $C$ and recover the exact median.

**Main theorem chain**

One run takes $O(n)$ and satisfies $\Pr[\textsc{Fail}] \leq n^{-1/4}$. Hence repeated independent restarts give a Las Vegas algorithm with linear expected running time.

# Takeaways

- Guess a structure cheaply;
- Verify it deterministically;
- Retry if needed.

# Exercise: Analyze the revised version of RMA

**Input / output**

Input: a set $S$ of $n$ elements over a totally ordered universe.
Output: the median element of $S$, denoted by $m$.

1. Pick a multiset $R$ of $n^{5/6}$ elements from $S$, independently and uniformly at random with replacement.
2. Sort $R$.
3. Let $d$ be the $\left(\frac{1}{2}n^{5/6} - n^{2/3}\right)$th smallest element of $R$.
4. Let $u$ be the $\left(\frac{1}{2}n^{5/6} + n^{2/3}\right)$th smallest element of $R$.
5. Compare every element of $S$ with $d$ and $u$ to compute
   $$C = \{x \in S : d \le x \le u\}, \quad \ell_d = \#\{x \in S : x < d\}, \quad \ell_u = \#\{x \in S : x > u\}.$$
6. If $\ell_d > n/2$ or $\ell_u > n/2$, output $\textsc{Fail}$.
7. If $|C| \le 4n^{5/6}$, sort $C$; otherwise output $\textsc{Fail}$.
8. Output the $\left(\lceil n/2 \rceil - \ell_d + 1\right)$th smallest element of $C$.

# Discussions